SOLUTION REQUIREMENTS

Date	07/07/2026	**Team ID**	XXXXXX
Project Name	EduGenie: Google Gemini-Powered Learning Assistant	**Maximum Marks**	4 Marks

Define Functional and Non-Functional Requirements:
Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.
• **Functional Requirements (FR):** Define specific behaviors, functions, or features of the system (What the system should do).
• **Non-Functional Requirements (NFR):** Specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/ Medium/ Low)
1	**Authentication**	The system must securely authenticate users via Google OAuth 2.0, supporting student, educator, and administrator sign-ins. It must implement secure session management with automatic JWT token expiration and secure cookie storage.	**High**
2	**Authorization levels**	Define RBAC (Role-Based Access Control): • **Student:** Access personalized learning paths, interactive quizzes, Gemini chatbot assistant, and progress dashboards. • **Educator:** Create and distribute curriculum content, monitor class analytics, override grading, and manage prompt baselines. • **Admin:** System settings, API quota limits, and user provisioning.	**High**
3	**External interfaces**	The system must interface directly with the **Google Gemini API** (Gemini 1.5 Pro / Flash models) for chat and content generation. It must sync seamlessly with Learning Management Systems (LMS) like Google Classroom using LTI standards, and connect to cloud databases (Firebase Cloud Firestore) for user history.	**High**
4	**Transactions processing**	Handles processing of LLM inference requests, conversational stream generation, multi-turn dialogue histories, interactive quizzes execution, and transactional saving of user milestones. It must efficiently queue streaming token payloads back to the client interface.	**Medium**
5	**Reporting**	Generate live visual dashboards displaying personalized analytics: student comprehension matrices, course completion	**Medium**

S.No	Requirement Category	Requirement Description	Priority (High/ Medium/ Low)
		rates, common conceptual bottlenecks identified by Gemini, and exportable weekly CSV progress summaries for educators.	
6	**Business rules, etc.**	• Educational content generated by Gemini must undergo structured context filtering against the assigned school syllabus. • Chat responses must restrict code or text generation to age-appropriate levels. • Daily message allowances per student must be enforced to manage API costs.	**High**
7	**Compliance to laws or regulations**	The platform must strictly comply with data privacy regulations including COPPA (Children's Online Privacy Protection Act), FERPA (Family Educational Rights and Privacy Act), and GDPR regarding student information tracking and AI data minimization rules.	**High**
8	**Other**	Provide a robust offline caching framework for completed study modules and allow speech-to-text input functionality for younger learners navigating the conversational system.	**Low**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	The UI must remain responsive. Gemini model streaming responses should start displaying tokens promptly after user input to simulate live engagement.	• Page loads in < 1.5 seconds. • Time-to-first-token (TTFT) for AI responses < 800ms.
2	**Scalability**	The architecture must dynamically scale to accommodate traffic surges during school hours and exam seasons without service disruption.	• Supports up to 25,000 concurrent active users. • Database horizontally scales seamlessly.
3	**Security & Data Privacy**	All information stored and transmitted must be securely locked to ensure privacy and integrity of user profiles and chat logs.	• End-to-end AES-256 encryption for data at rest. • TLS 1.3 for data in transit.
4	**Reliability & Availability**	The system must guarantee continuous availability and fail gracefully if external AI services face rate-limits or short outages.	• 99.95% uptime per month. • Automatic fallback caching or graceful queuing during Gemini API outages.
5	**Usability & Accessibility**	The application must offer a clean, clear, and distraction-free interface built for diverse learner capabilities.	• Complies with WCAG 2.1 AA accessibility standards. • Screen reader friendly.
6	**Other**		

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
		Maintainability and expandability must be prioritized to ensure upcoming Gemini model versions can be quickly adopted.	• Modular service-oriented design allowing LLM model swaps within 1 sprint.